

Lab Num: Two

**Topic: Creating a Virtual SD Card for
Embedded Linux Development**

- 1- Create a 1 GiB Virtual Disk Image and explain the command you use.

Exp.Output:

```
1023+0 records in
1023+0 records out
1072693248 bytes (1.1 GB, 1023 MiB) copied, 0.659766 s, 1.6 GB/s
```

- 2- Define what is the difference between the **DOS/MBR** and **GPT** partition scheme/type.
- 3- Define what is the difference between different File systems and Its usage (FAT16, FAT32, and EXT4)
- 4- Formatting and Partition the Virtual Disk Image.
 - a. First primary partition (200M + bootable + FAT16)
 - b. Second primary partition (rest of the available space + EXT4)

Exp.Output:

```

Disk: sd.img
Size: 1023 MiB, 1072693248 bytes, 2095104 sectors
Label: dos, identifier: 0xe3ab26ba

```

Device	Boot	Start	End	Sectors	Size	Id Type
sd.img1	*	2048	411647	409600	200M	6 FAT16
>> sd.img2		411648	2095103	1683456	822M	83 Linux

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- 5- Define what is the Loop Devices, why Linux use them.
 - a. Command to create a loop device
 - b. Command to list all loop devices currently in use.
 - c. Command to detach a (Mounted)loop device
- 6- How can you check the current loop device limit?
- 7- Can you expand the number of loop devices in Linux ?
- 8- Attach the Virtual Disk Image as a Loop Device.

Exp.Output:

```
loop36      7:36    0 1023M 0 loop
└─loop36p1  259:8     0 200M 0 part
└─loop36p2  259:9     0 822M 0 part
```

- 9- Format the Virtual Disk Image Partitions “Explain the command you use”
 - a. First primary partition (Boot partition → FAT16, label "boot")
 - b. Second primary partition (ext4, label "rootfs")
- 10- Explain what you Know about the “**mount**” and “**unmount**” Linux Command.
- 11- What is the difference between the block device vs character device.
- 12- Create Mount Points and Mount the Virtual Disk Image Partitions.

Exp.Output:

```
loop36      7:36    0 1023M 0 loop
└─loop36p1  259:8     0 200M 0 part /home/awadin/ITI_9Monthes/EmbeddedLinux_Kitchen/VirtualSdCard/boot
└─loop36p2  259:9     0 822M 0 part /home/awadin/ITI_9Monthes/EmbeddedLinux_Kitchen/VirtualSdCard/rootfs
```